

Inertia as a Spectral Separator: Feedback-Sector Hybridization in Low-Inertia Inverter-Dominated Power Networks

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Abstract—Low-inertia studies usually interpret inertia as an energy or frequency-security resource: less inertia increases RoCoF and larger virtual inertia is therefore often viewed as beneficial. This paper develops a complementary interpretation: inertia is also a *spectral separator* between network-dominated modes and converter-control modes. In an inverter-rich grid, reducing inertia migrates electromechanical modes upward in frequency, potentially bringing them into the bands occupied by phase-locked loops (PLLs), grid-following controls, grid-forming loops, filters, and other controller poles. We formulate this interaction after a controller-aware port reduction and an algebraic decomposition into feedback sectors. For each sector, a Sylvester separation measure quantifies the distance between the grid and controller invariant subspaces relative to their coupling. We prove a subspace-mixing bound, derive a closed-form inertia-resonance threshold under uniform inertia scaling, and obtain a two-mode hybridization law that predicts mode exchange near a controller/network encounter. The framework implies that total inertia alone cannot order stability: equal total inertia can yield different projected inertia in the critical feedback sector, and local virtual inertia can be non-monotone because it can first move a grid mode toward a controller resonance before detuning it away. An AC-consistent IEEE 39-bus reduced SG/GFL study provides falsification-oriented evidence. At fixed total synchronous inertia equal to 30% of the nominal effective value, 1,200 random placements span critical spectral abscissas from -0.326 to $+1.151 \text{ s}^{-1}$, with 316 stable and 884 unstable cases. In a separate low-inertia stress case, increasing inertia at a single synchronous bus produces a re-entrant stability window: the system is stable, becomes unstable for an intermediate inertia interval, and becomes stable again. At a uniform 25% inertia level, a small PLL retuning restores stability without restoring the missing inertia. These results support a planning principle: low-inertia remediation should preserve controller-network spectral separation inside the relevant feedback sectors, not merely maximize total inertia.

Index Terms—Low-inertia power systems, inverter-dominated grids, virtual inertia, phase-locked loop, modal interaction, spectral separation, Sylvester equation, grid-following inverter, model reduction, controller-network interaction.

I. INTRODUCTION

LOW-inertia power systems are commonly analyzed through frequency nadir, RoCoF, equivalent inertia, damping, and small-signal eigenvalues. These quantities remain essential, but converter-rich systems add a second difficulty: their controller dynamics occupy time scales that increasingly

overlap with network and electromechanical dynamics. Recent work has documented increased model stiffness in low-inertia systems and the resulting difficulty of preserving a clean slow/fast decomposition [1]. At the device level, PLL and virtual-inertia loops can interact adversely, and detailed studies report mode conversion, multi-timescale coupling, and stability trade-offs when virtual inertia is added to grid-following generation [2]–[5].

These observations suggest a question that is different from the usual “how much inertia is enough?”:

How much inertia, placed where, is required to keep network-dominated modes spectrally separated from controller-dominated modes?

The distinction matters because a controller/network encounter need not be visible in a scalar connectivity measure. Generalized-Fiedler metrics now combine topology and nodal inertia to characterize dynamic connectivity [6], while complex-frequency theory provides a geometric interpretation of linear modes [7]. Those developments are complementary to the present work but do not answer the sector-level time-scale separation problem considered here.

The central thesis of this paper is

inertia is both stored kinetic energy and a spectral-separation resource. (1)

The proposed framework has four contributions.

- C1. Feedback-sector separation.** A controller-aware closed-loop model is partitioned into network and controller subspaces inside algebraically meaningful feedback sectors. A Sylvester separation margin quantifies whether those subspaces can remain weakly mixed.
- C2. Inertia-resonance theory.** We derive the $\eta^{-1/2}$ migration law under uniform inertia scaling, a closed-form controller-resonant inertia threshold, and a two-mode hybridization formula that explains mode exchange and loss of model separation.
- C3. Inertia geometry rather than inertia quantity.** We show numerically that equal total inertia can yield stable or unstable outcomes depending only on placement. A local inertia sweep further exhibits a re-entrant stable–unstable–stable pattern.
- C4. Sparse inertia/controller co-design.** The theory yields a planning problem that can choose between virtual inertia placement and controller retuning. In the IEEE 39-bus reduced stress case, small PLL retuning restores stability at

a low-inertia operating point without replacing the missing inertia.

The paper does *not* claim that modal proximity itself is new. Open-loop modal proximity and multi-timescale interaction are active topics in converter stability [4]. The contribution pursued here is a network-level, feedback-sector formulation that makes inertia placement and controller retuning comparable through the same spectral-separation object.

II. CONTROLLER-AWARE FEEDBACK SECTORS

A. Reduced closed-loop operator

Consider a linearization of a nonlinear AC-DAE around an operating point. Retain a set of terminal/network states and eliminate internal controller states by a Schur complement. The reduced nonlinear eigenvalue problem can be written

$$F(s)x = 0, \quad F(s) = s^2M + sD + K + \Pi(s), \quad (2)$$

where K is the tangent electrical stiffness and $\Pi(s)$ is the frequency-dependent self-energy induced by eliminated controller/filter states. The poles of $\Pi(s)$ therefore retain controller memory even when the retained model is compact.

The cycle-algebra viewpoint developed in the companion scalarizability framework motivates a decomposition of the retained port space into invariant or approximately invariant closed-loop feedback sectors

$$\mathcal{V} = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_R. \quad (3)$$

Within one sector, choose coordinates that separate network-dominated and controller-dominated states:

$$A_r = \begin{bmatrix} A_{g,r}(M) & B_r \\ C_r & A_{c,r} \end{bmatrix}. \quad (4)$$

The sector formulation is important: a network can be globally high order while the dangerous controller/network encounter lives in a low-dimensional irreducible feedback block.

B. Sylvester separation

For square matrices A and B , define the classical separation

$$\text{sep}(A, B) = \sigma_{\min}(I \otimes A - B^T \otimes I). \quad (5)$$

This quantity controls the conditioning of the Sylvester equation $AX - XB = C$ [8]. To compare separation with the strength of the network/controller coupling, define

$$\mathfrak{S}_r = \frac{\text{sep}_G(A_{g,r}, A_{c,r})}{\sqrt{\|B_r\|_G \|C_r\|_G}} \quad (6)$$

using a physically meaningful metric G when non-unitary coordinate changes must be allowed.

Theorem 1 (Sector mixing bound). *Assume $\text{sep}(A_g, A_c) > 0$. For a first-order block-diagonalizing transformation with graph matrix X satisfying*

$$A_g X - X A_c = -B, \quad (7)$$

one has

$$\|X\|_F \leq \frac{\|B\|_F}{\text{sep}(A_g, A_c)}. \quad (8)$$

The analogous controller-to-grid correction satisfies $\|Y\|_F \leq \|C\|_F / \text{sep}(A_c, A_g)$.

Proof. Vectorization gives

$$(I \otimes A_g - A_c^T \otimes I) \text{vec}(X) = -\text{vec}(B). \quad (9)$$

Taking the smallest singular value of the Sylvester operator yields (8). The second inequality is identical with the roles reversed. $\square$

Remark 1. A small $\mathfrak{S}_r$ is not itself a stability certificate. It states that an invariant-subspace separation used by reduced or decoupled models is ill-conditioned relative to the coupling. Stability must still be computed from the full sector.

III. INERTIA AS A SPECTRAL SEPARATOR

A. Uniform inertia migration

Consider first the conservative sector eigenproblem

$$K_r v_r = \omega_r^2 M_r v_r. \quad (10)$$

Uniformly scale the inertia inside the sector as

$$M_r(\eta) = \eta M_r^0, \quad \eta > 0. \quad (11)$$

Theorem 2 (Inertia migration law). *If ω_{r0} is a nonzero natural frequency at $\eta = 1$, then*

$$\omega_r(\eta) = \frac{\omega_{r0}}{\sqrt{\eta}}. \quad (12)$$

Proof. Substitution into (10) gives $K_r v_r = \omega_r^2 \eta M_r^0 v_r$. Comparing with the $\eta = 1$ generalized eigenproblem gives $\omega_r^2 = \omega_{r0}^2 / \eta$. $\square$

Thus low inertia sweeps network modes upward through the controller spectrum. If a controller pole is

$$\mu_\ell = -\sigma_\ell + j\omega_{c,\ell}, \quad (13)$$

the simplest frequency-encounter threshold is

$$\eta_{r\ell}^* = \left(\frac{\omega_{r0}}{\omega_{c,\ell}} \right)^2. \quad (14)$$

Equation (14) is not a stability boundary by itself; it identifies the inertia level at which detuning vanishes in the conservative approximation.

B. Distributed inertia-resonance manifold

For heterogeneous inertia $M(\mathbf{m}) = \text{diag}(m_1, \dots, m_n)$, project onto sector r and define the controller-resonance manifold

$$\mathcal{R}_{r\ell} = \{ \mathbf{m} : \det(K_r - \omega_{c,\ell}^2 M_r(\mathbf{m})) = 0 \}. \quad (15)$$

This formulation makes explicit why the *location* of inertia matters: two inertia vectors with the same sum need not have the same projection onto the critical sector.

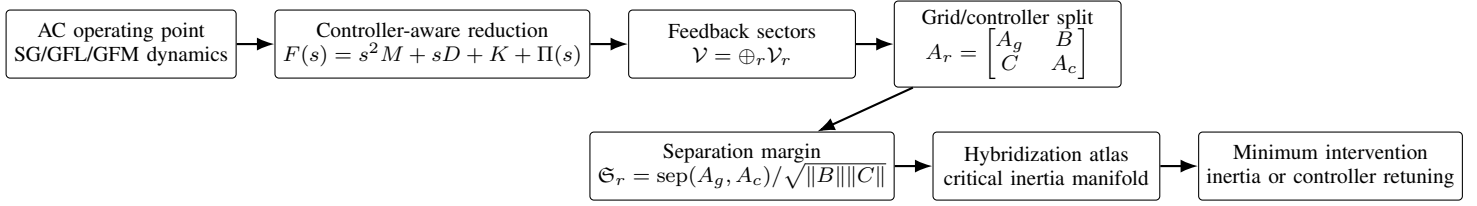


Fig. 1. Proposed low-inertia analysis. The key object is not total inertia but spectral separation between network and controller dynamics inside the relevant closed-loop feedback sectors.

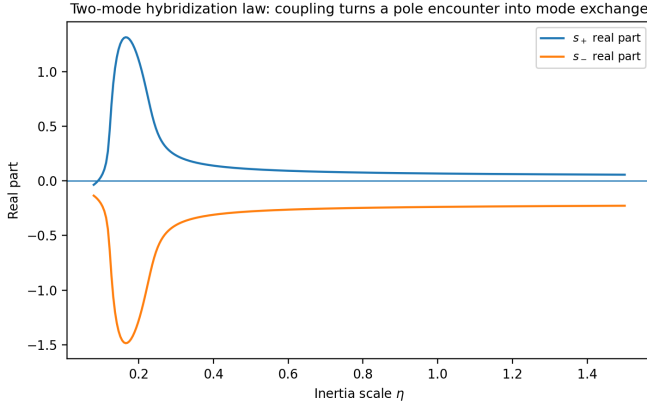


Fig. 2. Illustrative two-mode hybridization law (17). Coupling converts a grid/controller pole encounter into a mode exchange; the figure is conceptual, not IEEE 39-bus data.

C. Two-mode hybridization

Near an isolated encounter, project the full problem onto one network mode and one controller mode. A minimal local model is

$$\begin{bmatrix} s - s_g(\eta) & -g \\ -h & s - \mu_c \end{bmatrix} z = 0. \quad (16)$$

Proposition 1 (Hybridized poles). *The two coupled poles are*

$$s_{\pm} = \frac{s_g + \mu_c}{2} \pm \sqrt{\left(\frac{s_g - \mu_c}{2}\right)^2 + gh}. \quad (17)$$

Proof. Set the determinant of (16) to zero and apply the quadratic formula. $\square$

Far from the encounter, $|s_g - \mu_c| \gg |gh|^{1/2}$ and the modes retain their grid/controller identities. Near the encounter, the square-root term is dominated by coupling and mode identity is exchanged. This is the mechanism behind the use of the term *hybridization*; the concept is related to modal-proximity analyses in converter systems, but here it is embedded into a network-level inertia-placement problem.

D. Inertia and damping are different steering directions

For the nonlinear eigenvalue problem (2), let x_k, y_k be right/left eigenvectors and

$$T_k = y_k^* F_s(\lambda_k) x_k. \quad (18)$$

For local nodal perturbations m_i, d_i ,

$$\frac{\partial \lambda_k}{\partial m_i} = -\frac{\lambda_k^2 y_k^* E_i x_k}{T_k}, \quad (19)$$

$$\frac{\partial \lambda_k}{\partial d_i} = -\frac{\lambda_k y_k^* E_i x_k}{T_k}. \quad (20)$$

Hence

$$\frac{\partial \lambda_k}{\partial m_i} = \lambda_k \frac{\partial \lambda_k}{\partial d_i}. \quad (21)$$

For an oscillatory mode, virtual inertia therefore rotates/scales the local damping sensitivity in the complex plane rather than duplicating it. This motivates two-parameter controller co-design instead of a single “add damping” rule.

IV. PLANNING AND CO-DESIGN IMPLICATIONS

A. Minimum separation inertia

A planning formulation can distinguish frequency-security requirements from spectral-separation requirements. Let $C_M(\mathbf{m})$ be an inertia-placement cost. Define

$$\begin{aligned} \min_{\mathbf{m}} \quad & C_M(\mathbf{m}) \\ \text{s.t.} \quad & \mathfrak{S}_r(\mathbf{m}, \boldsymbol{\kappa}) \geq \mathfrak{S}_{\min}, \quad r \in \mathcal{R}_c, \\ & \mathbf{m}_{\min} \leq \mathbf{m} \leq \mathbf{m}_{\max}, \end{aligned} \quad (22)$$

where $\mathcal{R}_c$ is a set of critical feedback sectors. Equation (22) defines a *spectral-separation inertia reserve*; it is different from an aggregate RoCoF constraint.

B. Inertia/controller detuning co-design

Because both network and controller poles are movable, a cheaper formulation is

$$\begin{aligned} \min_{\Delta \mathbf{m}, \Delta \boldsymbol{\kappa}} \quad & C_M(\Delta \mathbf{m}) + C_{\kappa}(\Delta \boldsymbol{\kappa}) \\ \text{s.t.} \quad & \text{sep}_G(A_{g,r}(\mathbf{m} + \Delta \mathbf{m}), A_{c,r}(\boldsymbol{\kappa} + \Delta \boldsymbol{\kappa})) \\ & \geq s_{\min}, \quad r \in \mathcal{R}_c. \end{aligned} \quad (23)$$

The optimizer can therefore choose between adding virtual inertia, changing controller bandwidth, changing PLL gains, or combining small changes. The industrial implication is direct: a low-inertia warning need not automatically imply a large BESS installation if a small controller detuning produces greater separation.

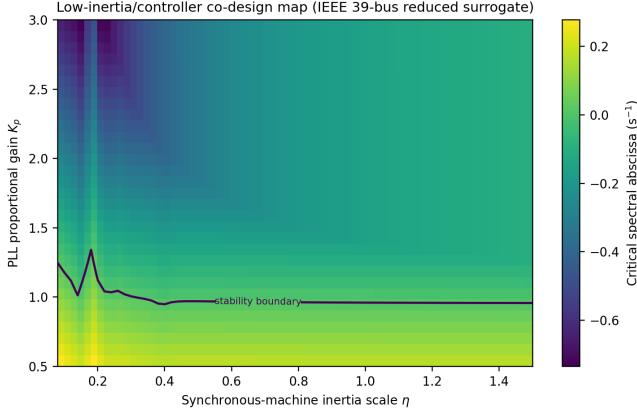


Fig. 3. Critical spectral abscissa as a function of synchronous-machine inertia scale and PLL proportional gain. The zero contour is the small-signal stability boundary. The map shows that inertia and controller tuning are coupled design variables.

V. IEEE 39-BUS FALSIFICATION STUDY

A. Model and scope

The numerical study uses the solved IEEE 39-bus AC operating point and reuses the reduced SG/GFL network model developed for the companion cycle-dynamics study. Ten generation ports are retained after Schur elimination of load buses. Six ports use classical swing/source dynamics and four ports (buses 30, 34, 37, and 38) use GFL models with PI PLLs and first-order P/Q current-control surrogates. The placement mirrors a common low-inertia conversion pattern in the IEEE 39-bus literature [10]. The present purpose is falsification of the low-inertia spectral-separation claims, not a claim of full-order EMT fidelity. MATPOWER provides the underlying benchmark data [11].

Unless otherwise stated, the stress study uses $K_i = 400$ and $K_p = 1$ for the four PLLs. This deliberately places the controller family near a low-damped region so that modal interaction can be observed. Negative-control sweeps with larger K_p remain stable over the same inertia range.

B. Uniform inertia reduction

All synchronous-machine inertias are scaled by η , leaving converter controls unchanged. Fig. 3 plots the critical spectral abscissa over the (η, K_p) plane. The stability boundary is not monotone in η : low inertia changes which grid mode is nearest to the PLL family, and controller retuning can move the boundary substantially without changing inertia.

At $K_p = 1$, the nearest decoupled grid/controller pole distance changes markedly as inertia is reduced while the critical full-order mode migrates through the same frequency band, as shown in Fig. 4. The intent of this plot is not to claim that Euclidean pole distance is a sufficient stability index; it visualizes the spectral encounter that motivates the sector separation measure.

C. Same total inertia, different stability

To test whether total inertia orders stability, the effective synchronous inertia budget was fixed at 30% of the nominal

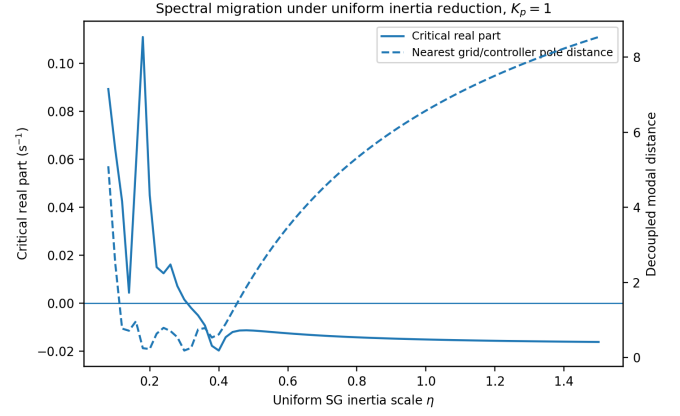


Fig. 4. Uniform inertia path for $K_p = 1$. Network-mode migration changes the nearest grid/controller pole distance while the full coupled spectral abscissa responds non-monotonically.

TABLE I
EQUAL-TOTAL-INERTIA PLACEMENT CAMPAIGN

Quantity	Result
Random placements	1,200
Stable placements	316
Unstable placements	884
Minimum critical real part	-0.326 s^{-1}
Median critical real part	$+0.084 \text{ s}^{-1}$
Maximum critical real part	$+1.151 \text{ s}^{-1}$
Total effective SG inertia	fixed at 30% nominal

value and randomly redistributed among the six synchronous generation buses. The budget is identical in all 1,200 trials; only placement changes. Table I summarizes the result.

Fig. 5 shows the distribution of critical abscissas and the most stable/unstable placements. The result is intentionally simple but important:

$$\sum_i M_i = \text{constant} \not\Rightarrow \alpha(A_{cl}) = \text{constant}. \quad (24)$$

Thus aggregate inertia does not determine controller/network separation.

D. A re-entrant inertia window

A stronger falsification test fixes all synchronous machines except bus 31 at 15% of nominal inertia and varies only the bus-31 inertia. The full system is stable for very small bus-31 inertia, becomes unstable over an intermediate interval, and becomes stable again for large bus-31 inertia. The numerically identified crossings are

$$H_{31}/H_{31}^{\text{nom}} \approx 0.1510 \quad \text{and} \quad 1.5169. \quad (25)$$

Inside the interval the critical mode has approximately 17 rad/s frequency, close to the weakly damped PLL band. Fig. 6 therefore provides a direct counterexample to a monotone “more local inertia is always better” interpretation in the coupled model.

This result should be interpreted narrowly. It is not a universal claim that inertia is destabilizing. It demonstrates that inertia

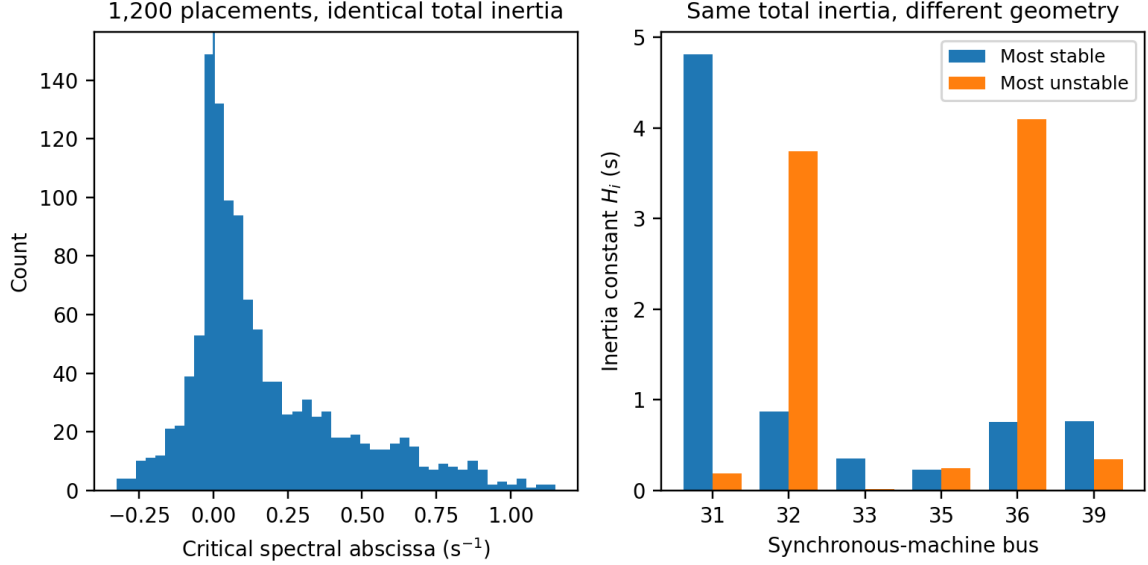


Fig. 5. Equal-total-inertia experiment. Left: distribution of critical spectral abscissa over 1,200 placements with identical total synchronous inertia. Right: the most stable and most unstable placements have the same total inertia but very different spatial distributions.

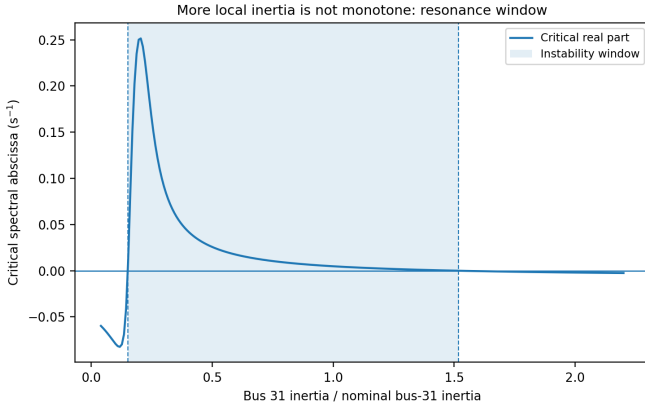


Fig. 6. Re-entrant stability under local inertia addition. All other synchronous machines are held at 15% of nominal inertia. Increasing bus-31 inertia first drives the controller-rich system into instability and only later restores stability.

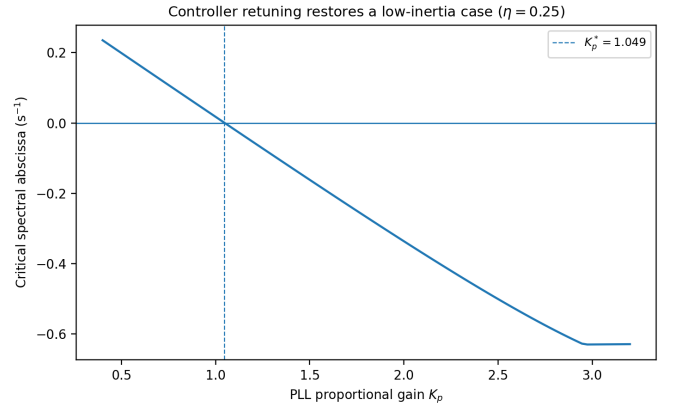


Fig. 7. At uniform inertia scale $\eta = 0.25$, small PLL retuning restores small-signal stability without increasing inertia. This motivates inertia/controller co-design rather than inertia-only remediation.

can be a *detuning parameter*: depending on the initial side of a controller/network encounter, moving a grid pole can initially decrease separation before eventually increasing it.

E. Controller retuning instead of inertia restoration

At a uniform 25% synchronous inertia level, the $K_p = 1$ stress case is unstable. Varying only the PLL proportional gain yields a stability threshold

$$K_p^* \approx 1.0486. \quad (26)$$

As Fig. 7 shows, a small controller change can recover stability without restoring the missing inertia. This does not imply that controller retuning substitutes for inertia in frequency-security constraints; it shows that the *small-signal modal problem* can be a detuning problem rather than an energy-quantity problem.

VI. RELATION TO EXISTING LOW-INERTIA METRICS

The proposed separation view is complementary to, not a replacement for, established metrics.

- **RoCoF and nadir** quantify frequency-security consequences of low kinetic energy. They remain necessary operational constraints.
- **Fiedler/GFV** quantify topology/inertia connectivity and frequency-strength structure [6]; they do not explicitly encode the controller pole spectrum.
- **Modal proximity and impedance methods** capture controller/network interaction at device or system level [2], [4]; the present formulation emphasizes sector-level separation and inertia placement.
- **Model-order reduction** addresses the numerical stiffness produced by overlapping time scales [1]. Here, the same loss of separation is treated as a planning/control phenomenon.

A practical workflow is therefore hierarchical: first enforce frequency-security inertia constraints; second identify the feedback sectors that support controller/network interaction; third enforce a minimum separation margin in those sectors; and finally choose the least-cost combination of inertia placement and controller tuning.

VII. OPEN THEORETICAL PROBLEMS AND FULL-ORDER VALIDATION

Several results remain to be established before the framework can support operational recommendations.

A. Exact feedback-sector extraction

The present IEEE 39-bus experiment uses the coarse grid/controller partition of the reduced state matrix. The next step is to compute the finest simultaneous invariant decomposition of the cycle-generated feedback algebra and evaluate $\mathfrak{S}_r$ only inside the irreducible sectors. This should sharpen the weak correlation observed when a global grid/controller separation is used indiscriminately.

B. Robust separation

Controller poles and inertia vary with operating point. A robust metric is

$$\mathfrak{S}_r^{\text{rob}} = \min_{p \in \mathcal{P}} \mathfrak{S}_r(p), \quad (27)$$

where $\mathcal{P}$ includes load, dispatch, controller uncertainty, topology, and IBR availability. The corresponding co-design problem can enforce separation over a family of operating points.

C. Full-order dynamic models

The strongest remaining validation gate is replication with detailed SG/GFL/GFM and converter inner-loop dynamics. The EPFL IEEE 39-bus benchmark provides low- and very-low-inertia configurations with inverter-connected generation [10]. Full-order Simulink/EMT or ANDES-type validation should test whether the re-entrant inertia window and equal-total-inertia placement reversal persist when current limits, DC-link dynamics, filters, and detailed converter controls are retained.

VIII. CONCLUSION

Low inertia changes more than the energy available to resist frequency acceleration. In a controller-rich network it also changes the relative location of network and controller spectra. This paper formalized that role through feedback-sector

separation, a Sylvester subspace-mixing bound, an inertia-resonance manifold, and a two-mode hybridization law. The reduced IEEE 39-bus experiments yielded three falsification-oriented findings: equal total inertia produced both stable and unstable placements, a local inertia increase created a re-entrant instability window, and small PLL retuning restored stability in a low-inertia case without restoring inertia. These results support a narrower but potentially important planning principle:

low-inertia adequacy is partly a spectral-placement problem, not only an (28)

The next step is to replace the coarse grid/controller partition by irreducible feedback-sector extraction and to repeat the experiments in full-order low-inertia benchmarks.

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